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HYDRO-MECHANICAL SOUNDING DEVICE FOR USE WITH ACOUSTIC TELEMETRY SYSTEM

Abstract

An acoustic telemetry system used with a wellbore assembly in a wellbore includes a hydro-mechanical sounding devices, relays, and a receiver. The sounding device includes a housing defining a chamber with an inlet port and at least one outlet port. A piston disposed in the chamber is movable against a strike face in the chamber in response to a hydraulic pressure differential. The impact of the piston against the strike face can produce an acoustic vibration, which can have defined characteristics. One or more relays disposed on the tubing electronically generate an acoustic signal in response to the acoustic vibration produced by the device. The acoustic signal can then be telemetered to a receiver at surface to determine a positive indication of an operation downhole.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Appl. No. 63/562,353, filed Mar. 7, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

[0002] Various types of data telemetry systems are used in wellbores. For example, mud pulse, wired drillpipe (WDP), or electromagnetic (EM) systems can be used. Many of these systems require complicated installation and high costs. For instance, current forms of acoustic telemetry require expensive electro-mechanical sounding technologies that use downhole batteries and electronics to create, communicate, and relay acoustic signals between various smart tools within a well completion.

[0003] For example, an electro-mechanical apparatus can be added to a tool to enable the tool to be used within an acoustic telemetry system. These “smart tools” can be actuated by these signals, or they can be actuated using typical hydraulic or mechanical means. Once actuated, these “smart tools” using the electro-mechanical apparatus can relay signals back to the surface operator to indicate that the smart tool has been functioned. It may be advantageous for very complex tools (e.g., downhole multi-function valves) that perform complex functions to have this level of technology and cost associated with them. However, many operations for downhole tool may be simpler in nature so adding complex electro-mechanical functions to such tools (e.g., packers and sleeves) can make an inexpensive tool cost prohibitive. With that said, many operators desire positive surface indications that these simpler downhole tools have functioned and completed their tasks.

[0004] The subject matter of the present disclosure is directed to overcoming, or at least reducing the effects of, one or more of the problems set forth above.

SUMMARY OF THE DISCLOSURE

[0005] A hydro-mechanical sounding device disclosed herein is used with a wellbore assembly in a wellbore. The hydro-mechanical sounding device comprises a housing and a piston. The housing defines a chamber therein. The housing defines at least one inlet port communicating the chamber with the wellbore assembly, and the housing further defines at least one outlet port communicating the chamber with the wellbore. The chamber has a striking surface toward the at least one outlet port. The piston is disposed in the chamber and has first and second faces. The first face is exposed toward the at least one inlet port of the chamber, and the second face is exposed toward the at least one outlet port of the chamber. The piston is movable from a first position proximate the at least one inlet port toward a second position against the striking surface in response to a hydraulic pressure differential between the first and second faces. The second face of the piston is configured to strike the striking surface and is configured to produce an acoustic vibration in response thereto.

[0006] An acoustic telemetry system disclosed herein is used with a wellbore assembly in a wellbore. The acoustic telemetry system comprises at least one sounding device, at least one relay, and a receiver. The at least one sounding device is disposed in the wellbore. The at least one sounding device comprises a piston and a striking surface disposed in a chamber. The piston is movable toward the striking surface in response to hydraulic pressure communicated from the wellbore assembly against the piston. In turn, the piston is configured to produce an acoustic vibration in response to striking against the striking surface. The at least one relay is disposed in the wellbore proximate the at least one sounding device. The at least one relay is configured to detect the acoustic vibration and is configured to electronically telemeter an acoustic signal in the

wellbore in response to the detection of the acoustic vibration. The receiver is configured to receive the acoustic signal telemetered from the at least one relay.

[0007] Each sounding device can be associated with an associated downhole tool disposed on the wellbore assembly. The piston can be movable in response to the hydraulic pressure produced by an operation of the associated downhole tool.

[0008] In one configuration, each relay comprises processing circuitry configured to: detect one or more characteristics of the acoustic vibration; and electronically produce, based on the one or more characteristics detected, a predefined one of a plurality of acoustic signals as a predefined acoustic signal. Additionally, the receiver comprises processing circuitry configured to: detect the predefined acoustic signal; and determine an indication of an event downhole based on the detection.

[0009] In another configuration, each relay is configured to receive the acoustic vibration and electronically produce the acoustic signal reproducing one or more characteristics of the acoustic vibration. Additionally, the receiver comprises processing circuitry configured to: detect the one or more characteristics of the acoustic signal; and determine an indication of an event downhole based on the detection.

[0010] A method disclosed herein is used with a wellbore assembly in a wellbore. The method comprises: performing an operation downhole; moving, in response to hydraulic pressure from the operation, a piston in a chamber of a hydro-mechanical sounding device disposed in the wellbore; producing an acoustic vibration by moving the piston in the chamber and/or striking a strike face of the piston moved against a striking surface of the chamber; receiving the acoustic vibration at a relay disposed in the wellbore proximate the hydro-mechanical sounding device; electronically producing an acoustic signal at the relay in response to the acoustic vibration received; telemetering the acoustic signal to a receiver; and determining, based on the acoustic signal telemetered thereto, an indication of the operation at the receiver.

[0011] The foregoing summary is not intended to summarize each potential embodiment or every aspect of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 illustrates a well completion having an acoustic telemetry system according to the present disclosure.

[0013] FIG. 2 illustrates a cross-sectional view of an example of a hydro-mechanical sounding device associated with a downhole tool.

[0014] FIG. 3A illustrates a cross-sectional view of a hydro-mechanical sounding device of the present disclosure in an unactuated state.

[0015] FIG. 3B illustrates a cross-sectional view of the hydro-mechanical sounding device of the present disclosure after actuation.

[0016] FIG. 3C illustrates a cross-sectional view of the hydro-mechanical sounding device of the present disclosure producing an acoustic signal.

[0017] FIG. 4 illustrates cross-sectional views of example strike plates for the hydro-mechanical sounding device of the present disclosure.

[0018] FIG. 5 illustrates cross-sectional views of example pistons for the hydro-mechanical sounding device of the present disclosure.

[0019] FIG. 6A illustrates a cross-sectional view of another hydro-mechanical sounding device of the present disclosure in an unactuated state.

[0020] FIG. 6B illustrates a cross-sectional view of yet another hydro-mechanical sounding device of the present disclosure in an unactuated state.

[0021] FIG. 6C illustrates a cross-sectional view of an additional hydro-mechanical sounding device of the present disclosure in an unactuated state.

[0022] FIG. 7A illustrates a cross-sectional view of a hydro-mechanical sounding device 50 of the present disclosure having consecutively freed pistons.

[0023] FIG. 7B illustrates a cross-sectional view of another hydro-mechanical sounding device 50 of the present disclosure having a modified surface in a chamber.

[0024] FIG. 7C illustrates a cross-sectional view of a combination hydro-mechanical sounding device 50 of the present disclosure.

[0025] FIG. 7D illustrates a cross-sectional view of another combination hydro-mechanical sounding device 50 of the present disclosure.

[0026] FIG. 8 illustrates a schematic view of an acoustic apparatus that can be used for a relay or a receiver of the disclosed acoustic telemetry system.

[0027] FIG. 9 illustrates a flow chart of a process according to the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0028] FIG. 1 illustrates a wellbore assembly 10 having an acoustic telemetry system 20 according to the present disclosure. The acoustic telemetry system 20 is used with tubing 14 in a wellbore 11 of the wellbore assembly 10. The acoustic telemetry system 20 includes a receiver 25, one or more relays or repeaters 30a-e, and one or more hydro-mechanical sounding devices 50a-c. In this example installation, five relays 30a-e and three hydro-mechanical sounding devices 50a-c are shown. Other installations can have different arrangements and number of elements.

[0029] The receiver 25, which can be at surface on a rig 12 or can be located elsewhere, has a receiver signal range (R1) and is configured to receive an acoustic input. The receiver 25 or other control unit 22 can then analyze the acoustic input to determine information according to the present disclosure, such as determining an indication of an operation performed downhole by a downhole tool.

[0030] The relays 30a-e are disposed on the tubing 14. Each relay 30a-e is configured to electronically produce an acoustic signal in response to an acoustic input received during use. To do this, each relay 30a-e may have a relay signal range (R2) to detect an acoustic input and to in turn telemeter the acoustic signal produced. As shown, each relay 30a-e is disposed on the tubing 14 within a relay signal range (R2) of at least one other of the relays 30a-e or within the receiver signal range (R1) of the receiver 25.

[0031] The acoustic telemetry devices 50a-c are hydro-mechanical sounding devices disposed on the tubing 14. When an operation is performed downhole, each of the hydro-mechanical sounding device 50a-c is configured to produce an acoustic vibration or signal in response to a change (e.g., increase) in hydraulic pressure in the tubing 14 from the performed operation. For example, a downhole tool 40a-c disposed on the tubing 14 proximate the hydro-mechanical sounding device 50a-c can be operated, which can generate the change in the hydraulic pressure to produce the acoustic vibration. Once generated, the acoustic vibration can travel through the walls of the tubing 14, through the fluid in the tubing 14, through the fluid in the wellbore annulus, or through a combination thereof.

[0032] The hydro-mechanical sounding device 50a-c can be implemented on or within a downhole tool 40a-c, such as a packer, a sleeve, a stage tool, or the like, to signal the occurrence of a downhole event. Alternatively, the hydro-mechanical sounding device 50a-c can be implemented as a stand-alone device to signal the occurrence of a downhole event. In general, the hydro-mechanical sounding device 50a-c can be mounted on the downhole tool 40a-c, can be integrated into housing components of the downhole tool 40a-c, can be mounted on a welded pup joint mount on the tubing 14, or can be mounted on the tubing 14 using brackets. These and other configurations can be used to incorporate the hydro-mechanical sounding device 50a-c with the wellbore assembly.

[0033] The hydro-mechanical sounding device 50a-c can be “tuned” to create an acoustic vibration

downhole with a predefined signature (e.g., a certain frequency, a decibel level, or the like) to signify a specific event in the operation of the well. To do this, the hydro-mechanical sounding device **50a-c** does not require the use of complex, expensive downhole batteries, or electro-mechanical equipment. Advantageously, the hydro-mechanical sounding device **50a-c** does not require temperature and pressure sensitive electronics so the hydro-mechanical sounding device **50a-c** can provide improved reliability for some implementations.

[0034] As noted, the hydro-mechanical sounding device **50a-c** can be “tuned” to create an acoustic vibration downhole having a predefined signature (e.g., a certain frequency, decibel level, or the like) to signify a specific event in the operation of the well. In conjunction with the actuation method (hydraulic or mechanical) of the downhole event and the timing of the downhole event, the hydro-mechanical sounding device **50a-c** can create an acoustic vibration within the wellbore that can be observed by the other acoustic telemetry relays **30a-e** within the well. In turn, these relays **30a-e** can provide strong positive feedback to an operator at surface that the event has occurred.

[0035] For example, if the specific event in the operation of the well is the functioning of a downhole tool **40a-c**, the hydro-mechanical sounding device **50a-c** can create an acoustic vibration within the wellbore **11** to indicate that the downhole tool **40a-c** has functioned as intended. If, for example, there are multiple sliding sleeves **40a-c** within the wellbore assembly **10**, a different “frequency” can be created or “tuned” independently for the operation of each sliding sleeve **40a-c**.

[0036] As shown in the example of FIG. 1, operation of the downhole tool **40a** can actuate the hydro-mechanical sounding device **50a** associated with the downhole tool **40a** to produce an acoustic vibration (**V1**). The relay **30c** within a relay signal range (**R2**) of the hydro-mechanical sounding device **50a** detects the acoustic vibration (**V1**) as acoustic input for the relay **30c**. The relay **30c** electronically produces an acoustic signal (**V2**) in response to the acoustic input from the sounding device **50a**. The relay **30c** has a relay signal range (**R2**) to telemeter the acoustic signal (**V2**) produced to another relay **30b**, which electronically produces an acoustic signal (**V3**) in response to the acoustic input from the relay **30c**. This relaying continues to telemeter a final acoustic signal (**V4**) to the receiver **25** for analysis.

[0037] As noted above, an acoustic signal can be relayed to surface to provide strong positive feedback to an operator at surface that a downhole event has occurred (e.g., a downhole tool **40a-c** has functioned as intended). Each acoustic signal can indicate which event has occurred (e.g., which downhole tool **40a-c** has functioned). Again, if there are multiple sliding sleeves **40a-c** within the wellbore assembly **10**, a different “frequency” can be created or “tuned” independently for the operation of each sliding sleeve **40a-c**.

[0038] FIG. 2 illustrates a cross-sectional view of an example of a hydro-mechanical sounding device **50** associated with a downhole tool, which is a sliding sleeve **40'** in this example. As shown, the sliding sleeve **40'** includes a housing **42** disposed on tubing **14**. An insert **46** is movable in the bore **44** of the housing **42**. For example, a plug or ball deployed to the sliding sleeve **40'** can land on a seat **47** of the insert **46**, and tubing pressure applied behind the seated ball can move the insert **46** in the bore **44**. Movement of the insert **46** can open the ports **48** in the housing **42** to communicate fluid to the wellbore annulus (not shown) outside the sliding sleeve **40'**.

[0039] The hydro-mechanical sounding device **50** can be attached to, incorporated into, or otherwise associated with the sliding sleeve **40'**. Briefly, the sounding device **50** includes a housing **52** having a chamber **54**, an inlet port **56**, and an outlet port **58**. A piston **60** is movably disposed in the chamber **54**. The chamber **54** can be filled with wellbore fluid with the outlet port **58** open to the wellbore annulus. Alternatively, chamber **54** can be filled with a predefined fluid with the outlet port **58** plugged with a temporary seal. Hydraulic pressure is communicated through the inlet port **56** to the chamber **54** (e.g., hydraulic pressure from one of the sliding sleeve's ports **48** can communicate with the device's inlet port **56**). The build-up of hydraulic pressure behind the piston **60** can shear it free so that the piston **60** moves in the chamber **54** toward the striking surface **72**. As the piston **60** moves, the fluid in the chamber **54** is expelled to the wellbore through the outlet

port **58**. Eventually, the piston's strike face **62** strikes a striking surface **72** of a strike plate **70** of the sounding device **50** with a strike force. The strike produces an acoustic vibration in the wellbore and/or the tubing **14**, which can then be detected as an acoustic input to a relay (**30**; FIG. **1**) within range. The expelled fluid from the outlet port **58** can also produce an acoustic vibration depending on the implementation.

[0040] Looking at the hydro-mechanical sounding device **50** in more detail, FIGS. **3A-3C** illustrates cross-sectional views of the disclosed sounding device **50** during operation. The hydro-mechanical sounding device **50** is shown in an unactuated state. As noted briefly above, the hydro-mechanical sounding device **50** includes a housing **52** and a piston **60**. The housing **52** defines a chamber **54** therein, and the housing **52** defines an inlet port **56** communicating the chamber **54** with tubing (e.g., a tubing bore, a tubing annulus, or a tool bore). The housing **52** also defines at least one outlet port **58** communicating the chamber **54** with the wellbore (e.g., wellbore annulus). The chamber **54** has a striking surface **72** toward the outlet port **58**. For assembly purposes, for example, a strike plate **70** or other component affixed to the housing **52** can include the striking surface **72** exposed to the chamber **54**.

[0041] The piston **60** is disposed in the chamber **54** and has first and second faces **62**, **63**. The first or piston face **63** is exposed toward the inlet port **56** of the chamber **54**, and the second or strike face **62** is exposed toward the outlet port **58** and the striking surface **72** of the chamber **54**.

[0042] As shown, a retainer or shear mechanism **66** temporarily retains the piston **60** in the unactuated position of FIG. **2**. The retainer **66**, which can be a shear pin, shear collar, or comparable element, is configured to release the piston **60** to move toward the strike position in response to a predetermined threshold of the hydraulic pressure differential in the chamber **54** acting on the piston **60**. As also shown, the piston **60** can have seals **64** that seal the piston **60** in the chamber **54**. When the pressure at the inlet port **56** is increased above the requisite shear value, the piston **60** is free to travel toward the strike plate **70**. When the strike plate **70** is struck, an acoustic vibration or sound is made that can then be picked up by the acoustic telemetry relays **30a-e** within the wellbore **11**.

[0043] The piston **60** is movable from a first position (FIG. **3A**) proximate the inlet port **56** toward a second position (FIG. **3C**) against the striking surface **72** in response to a hydraulic pressure differential between the piston's faces **62**, **63**. As shown in FIG. **3C**, the strike face **62** of the piston **60** is configured to strike the striking surface **72**, which produce an acoustic vibration.

[0044] One or more variables of at least one of the chamber **54**, the outlet port **58**, the striking surface **72**, the piston **60**, the strike face **62**, and the hydraulic pressure differential can be configured to tune one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device **50**. In general, the one or more characteristics of the acoustic vibration can be selected from the group consisting of amplitude, frequency, pitch, tone, etc.

[0045] One or more variables of at least one the striking surface **72**, chamber (**54**) surfaces, and the piston's strike face **62** can be configured to tune one or more characteristics of the acoustic vibration. For example, FIG. **4** illustrates cross-sectional views of example strike plates **70a-d** for the hydro-mechanical sounding device (**50**) of the present disclosure. Moreover, FIG. **5** illustrates cross-sectional views of example pistons **60a-c** for the hydro-mechanical sounding device (**50**) of the present disclosure.

[0046] In general, the strike plates **70a-d** and the pistons **60a-c** can be comprised of different materials. The materials can be the same or different from one another. The various materials for the strike plates **70a-d** and pistons **60a-c** can be used in various combinations to produce different acoustic characteristics.

[0047] Additionally, as shown in FIG. **4**, features of the striking surfaces **72a-d** can be configured to produce acoustic characteristics. Strike plate **70a** has a striking surface **72a** that is a flat surface. The striking surface **72b** on strike plate **70b** has a contoured surface **74**. The striking surface **72c** on strike plate **70c** has a collapsible surface **76**. Finally, the striking surface **72d** on strike plate **70d** has

a coated surface **78**, such as being coated with elastomer. One or more of these features can be combined with one another. These and other variations can be possible for the striking surfaces **72a-d**.

[0048] As shown in FIG. **5**, features of the piston's strike faces **62a-c** can be configured to produce acoustic characteristics. For example, the piston's strike face **62a** on the piston **60a** can have a flat surface. The piston's strike face **62b** on the piston **60c** can have a ridged surface, or the piston's strike face **62c** on the piston **60c** can have a contoured surface. These and other variations can be possible for the piston's strike face **62**. Additionally, any of the features of the strike faces **62a-c** used on the pistons **60a-c** can be used on the strike surfaces of the strike plate **70** and vis-a-versa. Overall, the examples of the strike faces **62a-c** and striking surfaces **72a-d** shown above can be interchanged and combined in different variations with one another to produce various characteristics. These and other alterations can be used for the strike faces **62a-c** and striking surfaces **72a-d**.

[0049] In addition to or in the alternative to strike and piston variables, one or more variables of the outlet port **58** of hydro-mechanical sounding device **50** can be configured to tune the one or more characteristics of the acoustic vibration. For example, FIGS. **6A-6C** illustrate cross-sectional views of other hydro-mechanical sounding devices **50** of the present disclosure. Examples of altered port size and positions are shown. The various geometries are interchangeable between the outlet ports **58**.

[0050] As shown in FIG. **6A**, an overall distance (X) of the piston **60** from the strike plate **70** can be configured to produce acoustic characteristics. As also shown in FIG. **6A**, the housing **52** can define a plurality of the outlet ports **58** disposed along a length of the chamber **54**. The one or more variables of the outlet ports **58** are configured to tune the one or more characteristics of the acoustic vibration. For example, spacing between the outlet ports **58** can be configured to tune the acoustic characteristics. As shown in FIG. **6A**, even spacing of the outlet ports **58** from one another along the length can be used. As shown in FIG. **6B**, varied spacing (Y) of the outlet ports **58** from one another along the length can be used. The different sizing and spacings can alter the duration, pulsing, and cadence of multiple sounds or notes in the acoustic vibration produced.

[0051] Size and/or shape of the outlet ports **58** can also be used to produce the acoustic characteristics. For example, equal sizing or different sizing of the outlet ports **58** can be used. Also, equal shaping or different shaping of the outlet ports **58** compared to one another can be used. For example, FIG. **6C** shows examples of alternative output ports **59** that operate as “whistles,” allowing expelled fluid passed through these outlet ports **59** to rapidly expand and oscillate as the fluid exits the outlet ports **59**. Various geometries can be used for these alternative outlet ports **59**.

[0052] To create and modulate the acoustic vibrations produced hydro-mechanically, the sounding device **50** can use other features as disclosed herein. For example, the sounding device **50** can use a shear mechanism **66** (i.e., shear pins **66**) having a predetermined threshold to retain the piston **60**, which is pinned and sized to strike the striking surface **72**, thus producing a sound. The piston **60** sheared free can move fluid within the volume of the chamber **54**. The moving fluid can provide a primary, secondary, or tertiary sound picked up by an acoustic telemetry relay. The chamber **54** can be ported such that the piston's movement is modulated, thus changing the manner or strike force with which the piston **60** strikes the striking surface **72** and subsequently defining the “tune” or frequency of the sound produced. The mating striking face **62** and striking surface **72** that contact one another may have interacting geometries to further modulate the frequency of the acoustic vibration produced. The outlet ports **58** may also be cut in such a way that a downhole “whistle” is created, further providing sound modulation to the sounding device **50**.

[0053] Previous examples include a strike plate **70** that may be a permanently fixed surface to be impacted by a single piston **60** traveling in the chamber **54**. In other configurations of the disclosed sounding device, multiple pistons **60** can be used in parallel to one another in different chamber

portions or in opposite directions within the same chamber. Also, multiple pistons **60** can be used in series, being arranged inline within the same chamber **54**. These configuration can induce multiple acoustic signals and sequential or consecutive shear events within the tool. For instance, multiple “pings” or other acoustic signals can be generated from the same device.

[0054] Turning then to FIG. 7A, a cross-sectional view of another hydro-mechanical sounding device **50** of the present disclosure shows multiple pistons **60a-n** arranged as a series in the chamber **54**. For this device **50** in FIG. 7A, several consecutive strike faces are provided within the chamber **54** using the plurality of pinned pistons **60a-n**.

[0055] During operations, an initial piston **60a** can be sheared free to move in the chamber **54**, and its strike face **62** can then engage a back face **63'** of a subsequent piston **60n** pinned in the chamber **54**. This initial impact can produce an initial acoustic signal. Pressure can eventually shear the subsequent piston **60n** free so that the pistons **60a-n** move in tandem along the chamber **54**. The tandem pistons **60a-n** can engage additional pistons (not shown) disposed in the chamber **54** to produce additional acoustic signals. Eventually, a final one of the pistons **60n** can eventually impact a striking surface **72** of a strike plate **70** to produce a final acoustic signal. The consecutive impacts of the freed pistons **60a-n** leading further into the chamber **54** can provide additional forms of variation to the acoustic signals disclosed herein. Several variations can be used for the piston's faces, shear values, separations between pistons **60a-n**, etc.

[0056] In a further configuration based on multiple outlet ports disposed along the chamber **54** as in FIGS. **6A-6C**, the arrangement shown here in FIG. 7A having multiple pistons **60a-n** may also have multiple outlet ports disposed along the chamber **54**. These outlet ports can be used to produce consecutive acoustic signals using the consecutively freed pistons **60a-n**.

[0057] FIG. 7B illustrate a cross-sectional view of yet another hydro-mechanical sounding device **50** of the present disclosure. Previous examples have included chambers **54** that may have a uniform surface. For this device **50** in FIG. 7B, the chamber **54** includes surface modifications **55** along the surface of the chamber **54**. These surface modifications **55** can include bumps, ridges, wickers, or other like profiles to modify the sound the piston **60** produces as it travels through the chamber **54**. In this way, the surfaces modifications **55** can be configured to tune the one or more characteristics of the acoustic vibration.

[0058] In previous examples, it may have been inferred that a given downhole tool (**40**) may have one sounding device **50** associated with the tool (**40**). This is not strictly necessary. A set of tools (**40**) can be associated with a shared sounding device **50**, or multiple sounding devices **50** can be associated with a given tool (**40**).

[0059] FIG. 7C illustrates a combination sounding device **50a** having a chamber with multiple chamber portions **54a-b** and having multiple pistons **60a-b**. Two chamber portions **54a-b** are shown, but more could be provided. Each chamber portion **54a-b** has its own piston **60-b**, but each can have more than one piston **60a-b**.

[0060] Each chamber portion **54a-b** can have its own inlet **56a-b**, which can communicate separately with the associated tool (**40**), although the inlets **56a-b** can be shared. Each chamber portion **54a-b** can have a different characteristic (e.g., diameter, length, size of outlet **58a-b**, number of outlets **58a-b**, etc.), and each piston **60a-b** can have a different characteristic (e.g., diameter, strike face, shear value, etc.). During operations, the combination sounding device **50a** can produce consecutive acoustic signals using the pistons **60a-b** if consecutively freed by predetermined thresholds of their retainers **68a-b**. For example, freeing one of the piston **60a** may be associated with a first operation of the associated tool (**40**), while freeing the other piston **60b** may be associated with a second operation of the associated tool (**40**). These and other variations are possible.

[0061] FIG. 7D illustrates another combination sounding device **50b** having multiple pistons **60a-b** in a shared chamber **54**. Each piston **60a-b** shares an inlet port **56**, which can communicate with the associated tool (**40**). Each side or portion of the chamber **54** can have a different characteristic (e.g.,

length, size of outlet **58a-b**, number of outlets **58a-b**, etc.), and each piston **60a-b** can have a different characteristic (e.g., diameter, strike face, shear value, etc.). During operations, the combination sounding device **50b** can produce a combined acoustic signal using the pistons **60a-b** associated with an operation of the associated tool (**40**). Alternatively, the combination sounding device **50b** can produce consecutive acoustic signals using the pistons **60a-b** if consecutively freed. For example, freeing one of the piston **60a** based on the shear value of the retainer **68a** may be associated with a first operation of the associated tool (**40**), while freeing the other piston **60b** based on the shear value of the retainer **68b** may be associated with a second operation of the associated tool (**40**). These and other variations are possible.

[0062] Returning to FIG. **1**, each tool **40a-c** or set of similar tools **40a-c** can be associated with a hydro-mechanical sounding device **50**, which can be “tuned” by one or more of the aforementioned arrangements. The sounding device **50** can therefore produce a characteristic sound to be measured by acoustic telemetry relays **30a-e** within the wellbore assembly **10**. Once the sound is picked up by these relays **30a-e**, the “message” associated with the produced acoustic signal can then be relayed to surface. At surface, the operator or software program can then decode this signal as a positive indication that a subsurface event associated with that frequency has occurred.

[0063] In the acoustic telemetry system **20**, the acoustic telemetry relays **30a-e** can be positioned at maximum extents of their signal range within the wellbore **11**. The downhole tools **40a-c** within the tubing **14** can each be equipped with a hydro-mechanical sounding device **50** that makes an acoustic vibration or noise that can be picked up and recognized by these relays **30a-e**. The acoustic input(s) from the relay(s) **30a-d** is then processed and relayed as acoustic signal(s) up the well to the operator at the surface.

[0064] Depending on a number of factors, the acoustic vibration or sound can be tuned by configuring one or a combination of the frequency, pitch, amplitude, decibel level, tone, etc. of the produced sound for a specific tool or shear event. Thus, each tool **40a-c** within the wellbore **11**, or a standalone sounding device **50**, can have its own signature sound for its actuation.

[0065] As noted above, several factors or inputs that may be configured or altered to affect the sound made by the sounding device **50**. For example, the distance (X) between the piston **60** and strike plate **70**; the shear value of the shear mechanism **66**; the shape and size of the inlet port **56**; the shape, size, number, and spacing of the outlet port(s) **58**; the number of pistons **60** (and their associated variables); and the number of chamber portions (and their associated) can be configured. In additional examples, the density of the fluid within the volume chamber **54** and the density of the fluid displacing the fluid within the volume chamber **54** can be configured. The surface finishes, surface coatings, contours, and the like on the strike face **62** and/or striking surface **72** can be configured. Surface finishes and coatings on the walls of the chamber **54** can be configured. Profiles, such as wickers, ridges, bumps, etc. can be configured on the surface of the chamber. The materials utilized for the piston **60**, strike plate **70**, and housing **52** can be configured. Additionally, the “sound(s)” created by the sounding device **50** can be produced by both the sounds made as the piston passes each outlet port **58** and the sound made once the strike plate **70** is hit.

[0066] In one configuration, each of the relays **30a-e** comprises processing circuitry that detects the acoustic vibration. Given that the relay **30a-e** is located or associated with a particular sounding device **50** or downhole tool **40** so the location or proximity of the acoustic vibration offers a discernible characteristic. The detection by the sounding device **50** already provides an indication of the associated event or operation that has occurred downhole. The relay's processing circuitry can electronically produce an acoustic signals, which can be indicative of the associated event, operation, location, source, or the like being monitored. In turn, processing circuitry of the receiver **25** or control unit **22** can detect the acoustic signal and determine the indication of the associated operation, location, source, or the like.

[0067] In another configuration, each of the relays **30a-e** comprises processing circuitry that detects one or more characteristics (e.g., frequency, amplitude, etc.) of the acoustic vibration. Based on the

one or more characteristics detected, the processing circuitry can electronically produce a predefined one of a plurality of acoustic signals, which is indicative of the associated event or operation being monitored. In turn, processing circuitry of the receiver **25** or control unit **22** detects the predefined acoustic signal and determines an indication of the associated event or operation based on the detection.

[0068] In yet another configuration, each of the relays **30a-e** can receive the acoustic vibration produced by the sounding device **50**. Processing circuitry of the relay **30a-e** can electronically generate an acoustic signal reproducing one or more characteristics of the original acoustic vibration. In turn, processing circuitry of the receiver **25** or control unit **22** can detect the one or more characteristics of the acoustic signal received. Using the detected characteristics, the processing circuitry can then determine an indication of the event.

[0069] FIG. **8** illustrates a schematic view of an acoustic apparatus **80** for the acoustic telemetry system. The acoustic apparatus **80** can be used for the relay (**30**) and the receiver (**25**) of the acoustic telemetry system (**20**). The acoustic apparatus **80** includes a bus **81**, a processing circuitry **82**, a memory or data storage component **83**, an acoustic telemetry unit **84**, an input/output communication interface **86**, and a power source or battery module **88**.

[0070] The bus **81** includes a component that permits communication among the components of the acoustic apparatus **80**. The processing circuitry **82** is implemented in hardware, firmware, or a combination of hardware and software. The processing circuitry **82** can be a microprocessor, a microcontroller, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or another type of processing component. In some examples, the processing circuitry **82** can include one or more processors capable of being programmed to perform a function.

[0071] The data storage component **83** may include one or more memories, such as a random-access memory (RAM), a read only memory (ROM), and/or another type of dynamic or static storage device (e.g., a flash memory, a magnetic memory, and/or an optical memory) that stores information and/or instructions for use by the processing circuitry **82**.

[0072] The input/output communication interface **86** includes a component that permits the acoustic apparatus **80** to send and receive signals and controls and to provide power to components. For example, the communication interface **86** may include an input component **87a** that receives sensor measurements, operational feedback, and other input information from the components of the completion. The communication interface **86** may also include an output component **87b** that provides control signals, actuations, and other output information from the acoustic apparatus **80** to the components of completion.

[0073] The acoustic telemetry unit **84** includes a transceiver-like component (e.g., a transceiver and/or a separate receiver **85a** and transmitter **85b**), which has one or more sensors, transducers, or other acoustic elements **87a-b**. The acoustic telemetry unit **84** enables the acoustic apparatus **80** to communicate acoustically with other equipment, such as with other relays, receivers, and the like. For example, the acoustic telemetry unit **84** permits the acoustic apparatus **80** to receive an acoustic signal from another relay (**30**) and from sounding devices **50** and to send an acoustic signal to another relay (**30**) or receiver (**25**).

[0074] The power source **88** is connected along the bus **81** to supply power to the processing circuitry **82** and other the internal components of the acoustic apparatus **80**. As a battery module, the power source **88** permits the acoustic apparatus **80** to be a portable integrated device for conducting the remote operations as disclosed herein.

[0075] The acoustic apparatus **80** may perform one or more processes described herein. The acoustic apparatus **80** may perform these processes by the processing circuitry **82** executing software instructions stored by a non-transitory computer-readable medium, such as the data storage component **83**. A computer-readable medium is defined herein as a non-transitory memory device. A memory device includes memory space within a single physical storage device or

memory space spread across multiple physical storage devices.

[0076] Instructions may be read into the data storage component **83** from another computer-readable medium or from another device via the acoustic telemetry unit **84**. When executed, instructions stored in the data storage component **83** may instruct the processing circuitry **82** to perform one or more processes described herein. Additionally, or alternatively, hardwired circuitry may be used in place of or in combination with instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0077] The number and arrangement of components shown in FIG. **8** are provided as an example. In practice, the acoustic apparatus **80** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **8**. Additionally, or alternatively, a set of components (e.g., one or more components) of the acoustic apparatus **80** may perform one or more functions described as being performed by another set of components of the acoustic apparatus **80**.

[0078] FIG. **9** is a flowchart of a process **100**, according to an example of the present disclosure. One or more process blocks of FIG. **9** may be performed by the acoustic telemetry system **20**, and reference numerals for the system **20** shown above are used in the discussion that follows.

[0079] The process **100** involves performing an operation downhole (block **102**). For example, a downhole tool (e.g., **40a** in FIG. **1**), such as a packer, a sleeve, a stage tool, or the like, disposed on the tubing **14** proximate the hydro-mechanical sounding device (e.g., **50a**) can be operated, which can generate the change in the hydraulic pressure to produce the acoustic vibration. The produced hydraulic pressure from the operation enters the chamber **54** of the proximate sounding device **50a** and moves the piston **60** in the chamber **54** of the sounding device **50a** (block **104**). As noted above, the piston **60** can be temporarily retained in place in the chamber **54** by a retainer or shear pin **66**. In response to a predetermined threshold of the hydraulic pressure, the piston **60** is released, and the hydraulic pressure forces the piston **60** in the chamber **54** toward the striking surface **72**. To better communicate the hydraulic pressure, the piston **60** can be sealed in the chamber **54** with seals **64**.

[0080] An acoustic vibration is then produced by moving of the piston **60** in the chamber **54** and/or by striking a strike face **62** of the piston **60** against the striking surface **72** of the chamber **54** (block **106**). In producing the acoustic vibration, one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device **50a** can be tuned by configuring one or more variables associated with at least one of: the chamber **54**, an outlet port **58** of the chamber **54**, the strike face **62**, the piston **60**, the striking surface **72**, the striking plate **70**, and the hydraulic pressure differential. In general and as noted above, the one or more configured variables can include values, parameters, or various details, such as: the length of the chamber **54**; the diameter of the chamber **54**; the thickness of the piston **60** and/or strike plate **70**; the number of outlet ports **58**; the material construction of the piston **60**, strike plate **70**, or the like; the strike force produced; surface design of the striking face **62** and/or striking surface **72**; etc. depending on the feature of the sounding device **50** in question.

[0081] Configuring the characteristics of the acoustic vibration can include configuring one or more of amplitude, frequency, tone, duration, cadence, and proximity of the acoustic vibration produced. For example, tuning the one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device **50a** may include defining a plurality of outlet ports **58** disposed along a length of the chamber **54**, and configuring, based on the outlet ports **58**, a strike force of the strike face **62** against the striking surface **72**. In another example, tuning the one or more characteristics of the acoustic vibration produced may include expelling fluid from an outlet port **58** of the chamber **54** in response to movement of the piston **60**; and producing the acoustic vibration with the expelled fluid. In yet another example, tuning the one or more characteristics of the acoustic vibration produced may include configuring the strike face **62** and/or the striking

surface **72**, such as by configuring the contour, shape, or material construction of the striking face **62** and/or the striking surface **72**.

[0082] Once generated, the acoustic vibration (e.g., **V1**) can travel through the walls of the tubing **14**, through the fluid in the tubing **14**, through the fluid in the wellbore annulus, or through a combination of these, until it reaches at least one proximate hydro-mechanical sounding device **50a-c**. The acoustic vibration **V1** is then received at a relay (e.g., **30c**) disposed in the wellbore proximate the actuated hydro-mechanical sounding device **50a** (block **108**), and the relay **30c** electronically produces an outgoing acoustic signal (e.g., **V2**) in response to the received acoustic vibration **V1** (block **110**). In one option, the relay **30** can act as a repeater in the sense that the relay **30** may reproduce the characteristics of the received acoustic vibration for transmission as the outgoing acoustic signal **V2**. In another option, the relay **30** may process the received acoustic vibration **V1** for the characteristics and may determine underlying information from the processing (i.e., determine the operation, the downhole tool **40a**, and/or other detail associated with the received acoustic vibration **V1**). In turn, the relay **30** may encode an indication of this underlying information in the outgoing acoustic signal **V2**.

[0083] Eventually, the acoustic signal is telemetered to a receiver **25** (block **112**). Depending on the location of the event downhole, several relays **30** may be needed to telemeter acoustic signals (e.g., **V2**, **V3**, **V4**) along the tubing string from the sounding device **50a** to the receiver **25**. In the end, the receiver **25** determines an indication of the operation based on the telemetered acoustic signal (block **114**). Operations at surface can automatically determine information about the operation downhole, which can be used to perform some additional automatic operation or may provide operators with information useful for further operational control.

[0084] Although FIG. **8** shows example blocks of the process **100**, the process **100** in some implementations may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **8**. Additionally, or alternatively, two or more of the blocks of the process **100** may be performed in parallel.

[0085] The acoustic telemetry system and methods disclosed here can be characterized by the following clauses:

[0086] Clause 1. A hydro-mechanical sounding device (**50**) used with a wellbore assembly (**10**) in a wellbore (**11**), the hydro-mechanical sounding device (**50**) comprising: [0087] a housing (**52**) defining a chamber (**54**) therein, the housing (**52**) defining at least one inlet port (**56**) communicating the chamber (**54**) with the wellbore assembly (**10**), the housing (**52**) defining at least one outlet port (**58**) communicating the chamber (**54**) with the wellbore, the chamber (**54**) having a striking surface (**72**) toward the at least one outlet port (**58**); and [0088] a piston (**60**) disposed in the chamber (**54**) and having first and second faces (**62**, **63**), the first face (**63**) exposed toward the at least one inlet port (**56**) of the chamber (**54**), the second face (**62**) exposed toward the at least one outlet port (**58**) of the chamber (**54**), the piston (**60**) being movable from a first position proximate the at least one inlet port (**56**) toward a second position against the striking surface (**72**) in response to a hydraulic pressure differential between the first and second faces, the second face of the piston (**60**) being configured to strike the striking surface (**72**) and configured to produce an acoustic vibration in response thereto.

[0089] Clause 2. The hydro-mechanical sounding device (**50**) of clause 1, comprising a retainer (**66**) temporarily retaining the piston (**60**) in the first position, the retainer (**66**) being configured to release the piston (**60**) to move toward the second position in response to a predetermined threshold of the hydraulic pressure differential.

[0090] Clause 3. The hydro-mechanical sounding device (**50**) of clause 1 or 2, wherein the piston (**60**) comprises seals sealing the piston (**60**) in the chamber (**54**).

[0091] Clause 4. The hydro-mechanical sounding device (**50**) of clause 1, 2 or 3, wherein one or more variables of at least one of the chamber (**54**), the outlet port (**58**), the strike face (**62**), the piston (**60**), the second face, and the hydraulic pressure differential are configured to tune one or

more characteristics of the acoustic vibration.

[0092] Clause 5. The hydro-mechanical sounding device (50) of clause 4, wherein the one or more characteristics of the acoustic vibration are selected from the group consisting of amplitude, frequency, tone, duration, pulsing, cadence, and proximity.

[0093] Clause 6. The hydro-mechanical sounding device of clause 4, wherein the chamber defines one or more modifications disposed on a surface of the chamber, the one or more modifications being configured to tune the one or more characteristics of the acoustic vibration.

[0094] Clause 7. The hydro-mechanical sounding device (50) of clause 4, 5 or 6, wherein the housing (52) defines a plurality of the at least one outlet port (58) disposed along a length of the chamber (54), the one or more variables of the outlet ports (58) being configured to tune the one or more characteristics of the acoustic vibration.

[0095] Clause 8. The hydro-mechanical sounding device (50) of clause 7, wherein the one or more variables of the outlet ports (58) comprises spacing between the outlet port (58) ports, the spacing including even spacing of the outlet ports (58) from one another along the length or varied spacing of the outlet ports (58) from one another along the length.

[0096] Clause 9. The hydro-mechanical sounding device (50) of clause 7 or 8, wherein the one or more variable of the outlet ports (58) comprises a size or a shape of the outlet ports (58), the size including equal sizing of the outlet ports (58) or different sizing of the outlet ports (58), the shape including equal shaping of the outlet ports (58) to one another or different shaping of the outlet ports (58) to one another.

[0097] Clause 10. The hydro-mechanical sounding device (50) of any one of clauses 4 to 9, wherein the one or more variables of at least one the strike face (62) and the second face are configured to tune one or more characteristics of the acoustic vibration.

[0098] Clause 11. The hydro-mechanical sounding device (50) of any one of clauses 1 to 10, wherein the strike face (62) comprises a first surface that is flat, contoured, collapsible, or coated; and wherein the second face comprises a second surface that is flat, ridged, or contoured.

[0099] Clause 12. The hydro-mechanical sounding device (50) of any one of clauses 1 to 11, wherein the housing (52) comprises a strike plate disposed in the chamber (54) and having the strike face (62).

[0100] Clause 13. The hydro-mechanical sounding device of any one of Clauses 1 to 13, wherein the piston is a final piston of a series; and wherein the series comprises one or more intermediate pistons disposed in the chamber between the at least one inlet and the final piston.

[0101] Clause 14. The hydro-mechanical sounding device of Clause 13, wherein a given one of the intermediate pistons is configured to strike the first face of the final piston or another one of the intermediate pistons and is configured to produce an intermediate acoustic vibration in response thereto.

[0102] Clause 15. The hydro-mechanical sounding device of Clause 1 to 14, wherein the chamber comprise a plurality of chamber portions, each of the chamber portions having one or more of the piston disposed therein.

[0103] Clause 16. The hydro-mechanical sounding device of Clause 15, wherein each of the chamber portions communicate with a separate one of the at least one inlet; or wherein each of the chamber portions communicate with a shared one of the at least one inlet.

[0104] Clause 17. An acoustic telemetry system used with a wellbore assembly (10) in a wellbore, the acoustic telemetry system comprising: at least one sounding device (50); at least one relay (30); and a receiver (25). For example, the acoustic telemetry system can comprise at least one sounding device (50) according to any one of claims 1 to 11 disposed in the wellbore. In another example, the at least one sounding device (50) disposed in the wellbore comprises a piston (60) and a striking surface (72) disposed in a chamber (54), the piston (60) being movable toward the striking surface (72) in response to hydraulic pressure communicated from the wellbore assembly (10) against the piston (60), the piston (60) being configured to produce an acoustic vibration in

response to striking against the striking surface (72).

[0105] The at least one relay (30) is disposed in the wellbore proximate the at least one sounding device (50). The at least one relay (30) is configured to detect the acoustic vibration and is configured to electronically telemeter an acoustic signal in the wellbore in response to the detection of the acoustic vibration. The receiver (25) is configured to receive the acoustic signal telemetered from the at least one relay (30).

[0106] Clause 18. The acoustic telemetry system of clause 17, wherein: the at least one relay (30) has a relay (30) signal range, the at least one relay (30) being disposed within the relay (30) signal range of the at least one sounding device (50) or at least one other of the at least one relay (30); and the receiver (25) has a receiver (25) signal range, the receiver (25) being disposed within the receiver (25) signal range of the at least one relay (30).

[0107] Clause 19. The acoustic telemetry system of clause 17 or 18, wherein each of the at least one sounding devices (50) is associated with a downhole tool disposed on the wellbore assembly (10), the piston (60) being movable in response to the hydraulic pressure produced by an operation of the associated downhole tool.

[0108] Clause 20. The acoustic telemetry system of clause 17, 18 or 19, wherein the at least one sounding device (50) comprises a housing (52) defining the chamber (54) therein, the housing (52) defining an inlet port (56) communicating the chamber (54) with the wellbore assembly (10), the housing (52) defining at least one outlet port (58) communicating the chamber (54) with the wellbore, the chamber (54) having the striking surface (72) toward the outlet port (58).

[0109] Clause 21. The acoustic telemetry system of clause 20, wherein the piston (60) is movably disposed in the chamber (54) and has a piston face (63) and a strike face (62), the piston face (63) exposed toward the inlet port (56) of the chamber (54), the strike face (62) exposed toward the at least one outlet port (58) of the chamber (54), the piston (60) being movable from a first position proximate the inlet port (56) toward a second position against the striking surface (72) in response to a differential of the hydraulic pressure between the piston face (63) and the strike face (62), the strike face (62) of the piston (60) being configured to strike the striking surface (72) and configured to produce the acoustic vibration in response thereto.

[0110] Clause 22. The acoustic telemetry system of clause 21, wherein the striking surface (72) is flat, contoured, collapsible, or coated; and wherein the strike face (62) of the piston (60) is flat, ridged, or contoured.

[0111] Clause 23. The acoustic telemetry system of clause 21 or 22, wherein the housing (52) comprises a strike plate disposed in the chamber (54) and having the striking surface (72).

[0112] Clause 24. The acoustic telemetry system of clause 21, 22 or 23, comprising a retainer (66) temporarily retaining the piston (60) in the first position, the retainer (66) being configured to release the piston (60) to move toward the second position in response to a predetermined threshold of the differential of the hydraulic pressure.

[0113] Clause 25. The acoustic telemetry system of any one of clauses 21 to 24, wherein the piston (60) comprises seals sealing the piston (60) in the chamber (54).

[0114] Clause 26. The acoustic telemetry system of clause 21, wherein one or more variables of at least one of the chamber (54), the outlet port (58), the strike face (62), the piston (60), the striking surface (72), and the hydraulic pressure differential are configured to tune one or more characteristics of the acoustic vibration.

[0115] Clause 27. The acoustic telemetry system of clause 26, wherein the one or more characteristics of the acoustic vibration are selected from the group consisting of amplitude, frequency, tone, duration, pulsing, cadence, and proximity.

[0116] Clause 28. The acoustic telemetry system of clause 26 or 27, wherein the housing (52) defines a plurality of the at least one outlet port (58) disposed along a length of the chamber (54), the one or more variables of the plurality of the outlet ports (58) being configured to tune the one or more characteristics of the acoustic vibration.

[0117] Clause 29. The acoustic telemetry system of clause 28, wherein the one or more variable of the plurality of the outlet ports (58) comprises spacing between the outlet ports (58), the spacing including even spacing of the outlet ports (58) from one another along the length or varied spacing of the outlet ports (58) from one another along the length.

[0118] Clause 30. The acoustic telemetry system of clause 28 or 29, wherein the one or more variable of the outlet ports (58) comprises a size or a shape of the outlet ports (58), the size including equal sizing of the outlet ports (58) or different sizing of the outlet ports (58), the shape including equal shaping of the outlet ports (58) to one another or different shaping of the outlet ports (58) to one another.

[0119] Clause 31. The acoustic telemetry system of any one of clauses 26 to 30, wherein the one or more variables of at least one the striking surface (72) and the strike face (62) are configured to tune the one or more characteristics of the acoustic vibration.

[0120] Clause 32. The acoustic telemetry system of any one of clauses 17 to 31, wherein: [0121] each of the at least one relay (30) comprises processing circuitry configured to: [0122] detect one or more characteristics of the acoustic vibration; and [0123] electronically produce, based on the one or more characteristics detected, a predefined one of a plurality of acoustic signals; and [0124] the receiver (25) comprises processing circuitry configured to: [0125] detect the predefined acoustic signal; and [0126] determine an indication of an event downhole based on the detection.

[0127] Clause 33. The acoustic telemetry system of any one of clauses 17 to 32, wherein: [0128] each of the at least one relay (30) is configured to receive the acoustic vibration and electronically produce the acoustic signal reproducing one or more characteristics of the acoustic vibration; and [0129] the receiver (25) comprises processing circuitry configured to: [0130] detect the one or more characteristics of the acoustic signal; and [0131] determine an indication of an event downhole based on the detection.

[0132] Clause 34. A method used with a wellbore assembly (10) in a wellbore, the method comprising: performing an operation downhole; moving, in response to hydraulic pressure from the operation, a piston (60) in a chamber (54) of a hydro-mechanical sounding device (50) disposed in the wellbore; producing an acoustic vibration by moving the piston in the chamber and/or striking a strike face (62) of the piston (60) moved against a striking surface (72) of the chamber (54); receiving the acoustic vibration at a relay (30) disposed in the wellbore proximate the hydro-mechanical sounding device (50); electronically producing an acoustic signal at the relay (30) in response to the acoustic vibration received; telemetering the acoustic signal to a receiver (25); and determining, based on the acoustic signal telemetered thereto, an indication of the operation at the receiver (25).

[0133] Clause 35. The method of clause 34, comprising temporarily retaining the piston (60) in a first position in the chamber (54); and releasing the piston (60) to move toward the striking surface (72) in response to a predetermined threshold of the hydraulic pressure.

[0134] Clause 36. The method of clause 34 or 35, comprising sealing the piston (60) in the chamber (54) with seals.

[0135] Clause 37. The method of clause 34, 35 or 36, wherein producing the acoustic vibration by striking the strike face (62) of the piston (60) moved against the striking surface (72) of the chamber (54) comprises tuning one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device (50).

[0136] Clause 38. The method of clause 37, wherein tuning the one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device (50) comprises configuring one or more variables of at least one of the chamber (54), an outlet port (58) of the chamber (54), the strike face (62), the piston (60), the striking surface (72), and the hydraulic pressure differential.

[0137] Clause 39. The method of clause 38, wherein the one or more characteristics of the acoustic vibration are selected from the group consisting of amplitude, frequency, tone, duration, cadence, and proximity.

[0138] Clause 40. The method of clause 37, 38 or 39, wherein tuning the one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device (50) comprises defining a plurality of outlet ports (58) disposed along a length of the chamber (54), and configuring, based on the outlet ports (58), a strike force of the piston face (63) against the striking surface (72).

[0139] Clause 41. The method of any one of clauses 37 to 40, wherein tuning the one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device (50) comprises expelling fluid from an outlet port (58) of the chamber (54) in response to movement of the piston (60); and producing the acoustic vibration with the expelled fluid.

[0140] Clause 42. The method of any one of clauses 37 to 41, wherein tuning the one or more characteristics of the acoustic vibration produced by the hydro-mechanical sounding device (50) comprises configuring the strike face (62) and the striking surface (72).

[0141] Clause 43. The method of any one of clauses 34 to 42, wherein performing the operation downhole comprises operating a downhole tool disposed on the wellbore assembly (10) proximate the hydro-mechanical sounding device (50).

[0142] The foregoing description of preferred and other embodiments is not intended to limit or restrict the scope or applicability of the inventive concepts conceived of by the Applicants. It will be appreciated with the benefit of the present disclosure that features described above in accordance with any embodiment or aspect of the disclosed subject matter can be utilized, either alone or in combination, with any other described feature, in any other embodiment or aspect of the disclosed subject matter.

[0143] In exchange for disclosing the inventive concepts contained herein, the Applicants desire all patent rights afforded by the appended claims. Therefore, it is intended that the appended claims include all modifications and alterations to the full extent that they come within the scope of the following claims or the equivalents thereof.

Claims

1. A hydro-mechanical sounding device used with a wellbore assembly in a wellbore, the hydro-mechanical sounding device comprising: a housing defining a chamber therein, the housing defining at least one inlet port communicating the chamber with the wellbore assembly, the housing defining outlet ports disposed along a length of the chamber, the outlet ports communicating the chamber with the wellbore, the chamber having a striking surface; and a piston disposed in the chamber and having first and second faces, the first face exposed toward the at least one inlet port of the chamber, the second face exposed toward the outlet ports of the chamber, the piston being movable from a first position proximate the at least one inlet port toward a second position against the striking surface in response to a hydraulic pressure differential between the first and second faces, the second face of the piston being configured to strike the striking surface and configured to produce an acoustic vibration in response thereto, wherein one or more first variables of the outlet ports are configured to tune one or more characteristics of the acoustic vibration.

2. The hydro-mechanical sounding device of claim 1, wherein the piston comprises one of: a retainer temporarily retaining the piston in the first position, the retainer being configured to release the piston to move toward the second position in response to a predetermined threshold of the hydraulic pressure differential; and seals sealing the piston in the chamber.

3. (canceled)

4. The hydro-mechanical sounding device of claim 1, wherein one or more second variables of at least one of the chamber, the striking surface, the piston, the second face, and the hydraulic pressure differential are configured to tune the one or more characteristics of the acoustic vibration.

5. The hydro-mechanical sounding device of claim 1, wherein the one or more characteristics of the acoustic vibration are selected from the group consisting of amplitude, frequency, tone, duration,

pulsing, cadence, and proximity.

6. The hydro-mechanical sounding device of claim 4, wherein one of: the chamber define one or more modifications disposed on a surface of the chamber, the one or more modifications as the one or more second variables being configured to tune the one or more characteristics of the acoustic vibration; and the one or more second variables of at least one of the striking surface and the second face are configured to tune the one or more characteristics of the acoustic vibration.

7. (canceled)

8. The hydro-mechanical sounding device of claim 1, wherein the one or more first variables of the outlet ports comprise at least one of: spacing between the outlet ports, the spacing including one of even spacing of the outlet ports from one another along the length and varied spacing of the outlet ports from one another along the length; a size of the outlet ports, the size including one of equal sizing of the outlet ports and different sizing of the outlet ports; and a shape of the outlet ports, the shape including one of equal shaping of the outlet ports to one another and different shaping of the outlet ports to one another.

9. (canceled)

10. The hydro-mechanical sounding device of claim 1, wherein one of: the striking surface comprises a first surface that is one of flat, contoured, collapsible, and coated; the second face comprises a second surface that is one of flat, ridged, and contoured; and the housing comprises a strike plate disposed in the chamber and having the striking surface.

11. (canceled)

12. The hydro-mechanical sounding device of claim 1, wherein the piston is a final piston of a series; and wherein the series comprises one or more intermediate pistons disposed in the chamber between the at least one inlet port and the final piston.

13. The hydro-mechanical sounding device of claim 12, wherein a given one of the intermediate pistons is configured to strike the first face of one of the final piston and another one of the intermediate pistons and is configured to produce an intermediate acoustic vibration in response thereto.

14. The hydro-mechanical sounding device of claim 1, wherein the chamber comprises a plurality of chamber portions, each of the chamber portions having one or more of the piston disposed therein.

15. The hydro-mechanical sounding device of claim 14, wherein one of: each of the chamber portions communicates with a separate one of the at least one inlet port; and each of the chamber portions communicates with a shared one of the at least one inlet port.

16. An acoustic telemetry system used with the wellbore assembly in the wellbore, the acoustic telemetry system comprising: at least one sounding device according to claim 1 disposed in the wellbore; at least one relay disposed in the wellbore proximate the at least one sounding device, the at least one relay being configured to detect the acoustic vibration and being configured to electronically telemeter an acoustic signal in the wellbore in response to the detection of the acoustic vibration; and a receiver being configured to receive the acoustic signal telemetered from the at least one relay.

17. The acoustic telemetry system of claim 16, wherein each of the at least one sounding device is associated with an associated downhole tool disposed on the wellbore assembly, the piston being movable in response to the hydraulic pressure differential produced by an operation of the associated downhole tool.

18. The acoustic telemetry system of claim 16, wherein one of: (i)(a) each of the at least one relay comprises processing circuitry configured to detect the one or more characteristics of the acoustic vibration and electronically produce, based on the one or more characteristics detected, a predefined one of a plurality of acoustic signals as a predefined acoustic signal; and (b) the receiver comprises processing circuitry configured to detect the predefined acoustic signal and determine an indication of an event downhole based on the detection; and (ii)(a) each of the at least one relay is

configured to receive the acoustic vibration and electronically produce the acoustic signal reproducing the one or more characteristics of the acoustic vibration, and (b) the receiver comprises processing circuitry configured to detect the one or more characteristics of the acoustic signal and determine the indication of the event downhole based on the detection.

19-20. (canceled)

21. A hydro-mechanical sounding device used with a wellbore assembly in a wellbore, the hydro-mechanical sounding device comprising: a housing defining a chamber therein, the housing defining at least one inlet port communicating the chamber with the wellbore assembly, the housing defining at least one outlet port communicating the chamber with the wellbore, the chamber having a striking surface toward the at least one outlet port; and a series of pistons disposed in the chamber, the pistons of the series comprising a final piston and comprising one or more intermediate pistons disposed in the chamber between the at least one inlet port and the final piston, each of the pistons having first and second faces, the first face exposed toward the at least one inlet port of the chamber, the second face exposed toward the at least one outlet port of the chamber, the final piston being movable from a first position proximate the at least one inlet port toward a second position against the striking surface in response to a hydraulic pressure differential between the first and second faces, the second face of the final piston being configured to strike the striking surface and configured to produce an acoustic vibration in response thereto.

22. The hydro-mechanical sounding device of claim 21, wherein the final piston comprises one of: a retainer temporarily retaining the final piston in the first position, the retainer being configured to release the final piston to move toward the second position in response to a predetermined threshold of the hydraulic pressure differential; and seals sealing the final piston in the chamber.

23. The hydro-mechanical sounding device of claim 21, wherein one or more variables of at least one of the chamber, the at least one outlet port, the striking surface, one or more of the pistons, the second face of one or more of the pistons, and the hydraulic pressure differential are configured to tune one or more characteristics of the acoustic vibration.

24. The hydro-mechanical sounding device of claim 23, wherein one of: the one or more characteristics of the acoustic vibration are selected from the group consisting of amplitude, frequency, tone, duration, pulsing, cadence, and proximity; the chamber defines one or more modifications disposed on a surface of the chamber, the one or more modifications as the one or more variables being configured to tune the one or more characteristics of the acoustic vibration; the housing defines a plurality of the at least one outlet port disposed along a length of the chamber, the one or more variables of the outlet ports being configured to tune the one or more characteristics of the acoustic vibration; and the one or more variables of at least one of the striking surface and the second face of the final piston are configured to tune the one or more characteristics of the acoustic vibration.

25. The hydro-mechanical sounding device of claim 21, wherein one of: the striking surface comprises a first surface that is one of flat, contoured, collapsible, and coated; the second face of the final piston comprises a second surface that is one of flat, ridged, and contoured; and the housing comprises a strike plate disposed in the chamber and having the striking surface.

26. The hydro-mechanical sounding device of claim 21, wherein a given one of the intermediate pistons is configured to strike the first face of one of the final piston and another one of the intermediate pistons and is configured to produce an intermediate acoustic vibration in response thereto.

27. The hydro-mechanical sounding device of claim 21, wherein the chamber comprise a plurality of chamber portions, each of the chamber portions having one or more of the pistons disposed therein.

28. The hydro-mechanical sounding device of claim 27, wherein one of: each of the chamber portions communicates with a separate one of the at least one inlet port; and each of the chamber portions communicates with a shared one of the at least one inlet port.

29. An acoustic telemetry system used with the wellbore assembly in the wellbore, the acoustic telemetry system comprising: at least one sounding device according to claim 21 disposed in the wellbore; at least one relay disposed in the wellbore proximate the at least one sounding device, the at least one relay being configured to detect the acoustic vibration and being configured to electronically telemeter an acoustic signal in the wellbore in response to the detection of the acoustic vibration; and a receiver being configured to receive the acoustic signal telemetered from the at least one relay.

30. The acoustic telemetry system of claim 29, wherein each of the at least one sounding device is associated with an associated downhole tool disposed on the wellbore assembly, the final piston being movable in response to the hydraulic pressure differential produced by an operation of the associated downhole tool.

31. The acoustic telemetry system of claim 29, wherein one of: (i))(a) each of the at least one relay comprises processing circuitry configured to detect one or more characteristics of the acoustic vibration and electronically produce, based on the one or more characteristics detected, a predefined one of a plurality of acoustic signals as a predefined acoustic signal; and (b) the receiver comprises processing circuitry configured to detect the predefined acoustic signal and determine an indication of an event downhole based on the detection; and (ii)(a) each of the at least one relay is configured to receive the acoustic vibration and electronically produce the acoustic signal reproducing the one or more characteristics of the acoustic vibration, and (b) the receiver comprises processing circuitry configured to detect the one or more characteristics of the acoustic signal and determine the indication of the event downhole based on the detection.
